

NEW LIFE RESEARCH FOUNDATION

2026 DATA SCIENCE AND AI SUMMER SCHOOL

PYTHON PROGRAMMING



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New Life Research Foundation (NELIREF) Ltd/Gte

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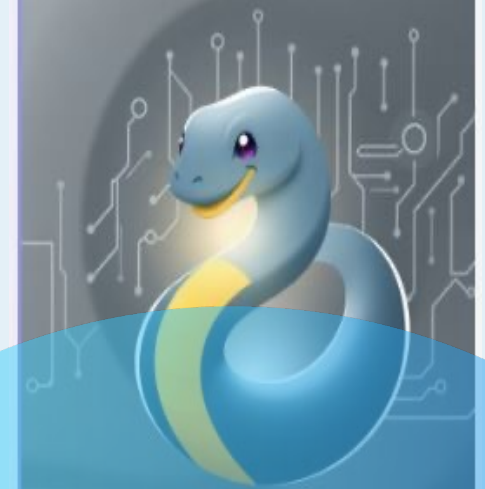
INTRODUCTION TO PYTHON PROGRAMMING

What is Python? Why Learn It?

Python is an interpreted, object-oriented, high-level programming language. It is a versatile, beginner-friendly programming language valued for its simplicity and readability. Its wide range of applications makes it an essential skill for modern Programmers.

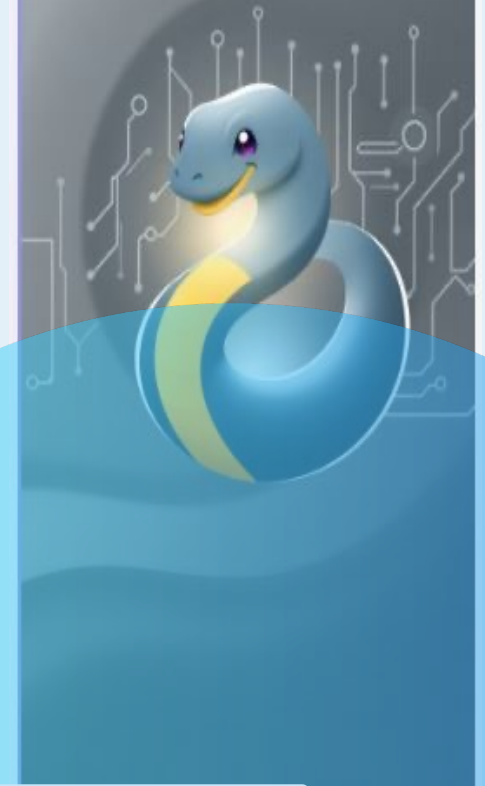
Installing Python & Setting Up

Learn to install Python on your computer and set up popular integrated development environments (IDEs) such as IDLE, VS Code, and Jupyter Notebook. Ensuring the development environment works smoothly is the first step to coding successfully.



Python Programming Areas of Application

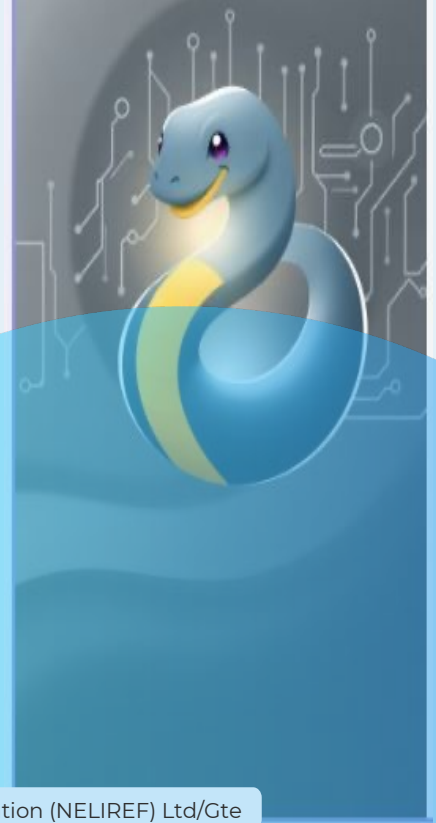
- ☐ Data visualization
- ☐ Data analysis
- ☐ Automation
- ☐ Artificial intelligence
- ☐ Software development
- ☐ Prototyping
- ☐ Web application
- ☐ Machine learning
- ☐ System scripting
- ☐ Workflows
- ☐ Mathematics
- ☐ Web development
(server side) and more



Python Programming

Areas of Application

- ❖ Data manipulation (e.g, libraries such as pandas, numpy, polars, etc)
- ❖ Data visualization (e.g, plotly, matplotlib, seaborn, etc)
- ❖ Statistical analysis (e.g, scipy, statsmodels, etc)
- ❖ Machine Learning (e.g, JAX, keras, theano, XGBoost, scikit learn, Tensorflow, PyTorch, etc)
- ❖ Natural Language Processing (NLP): E.g, NLTK, BERT, spaCy, Genism, Text Blob, Polyglot, pattern, etc
- ❖ Database operations (e.g, PySpark, Dask, Hodoop)
- ❖ Time series analysis (e.g, sktime, Darts, Prophet, tsfresh)
- ❖ Web scraping (e.g, Scrapy, beautiful soup, selenium)



Operators, Expressions, and Control Flow

Python Logic and Code Flow



Understanding Operators and Expressions

Master different types of operators in Python, such as arithmetic, comparison, logical, and assignment operators. Learn how to form expressions by combining operators and operands for calculations and logical evaluation.



Variables, Data Types, and I/O

Get familiar with how to work with variables and different data types in Python. Practice collecting user input, producing output, and writing readable code using comments.



Conditional Statements and Loops

Control the way your code executes using if, elif, and else statements. Harness the power of iteration through for loops, while loops, and even simulate dowhile loops. Utilize break, continue, and pass statements to refine your control flow.



Working with Python Data Structures

Core Data Handling



Lists, Tuples, Sets, and Dictionaries

Python offers several built-in data structures for storing and organizing data: lists for ordered collections, tuples for immutable sequences, sets for unique elements, and dictionaries for key-value pairs. Understanding these structures is fundamental to effective programming.

Accessing and Modifying Data

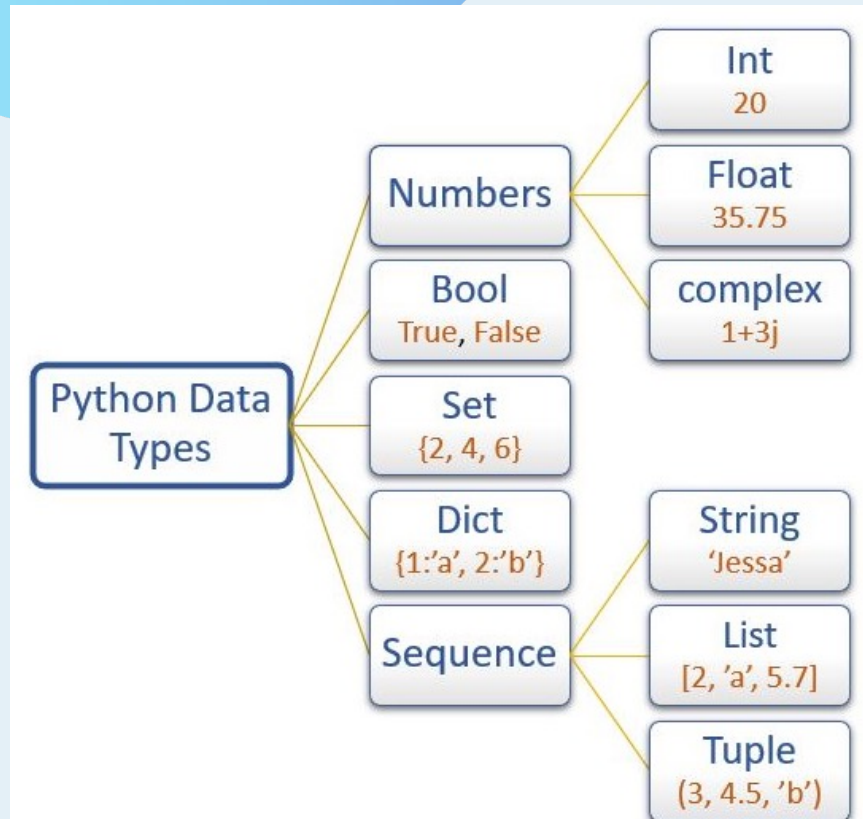
Learn how to access data using indexing and slicing, as well as how to add, remove, or update elements in each structure. Grasping these operations is essential for dynamic data manipulation in your programs.

Iterating Through Data

Explore techniques for looping through various data structures, which allows for efficient processing and transformation of stored data.

Working with Python Data Structures

Core Data Handling



Characteristics of Python Data Structures

LIST	TUPLE	DICTIONARY	SET
Allows duplicate members	Allows duplicate members	No duplicate members	No duplicate members
Changeable	Not changeable	Changeable indexed	Cannot be changed, but can be added, non-indexed
Ordered	Ordered	Unordered	Unordered
Square bracket []	Round brackets ()	Curly brackets { }	Curly brackets { }

Questions & Answers



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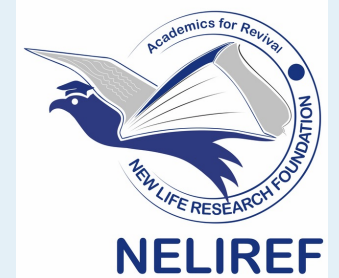
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We admonish you to live
for Jesus and connect
with our community!

Thank You!



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